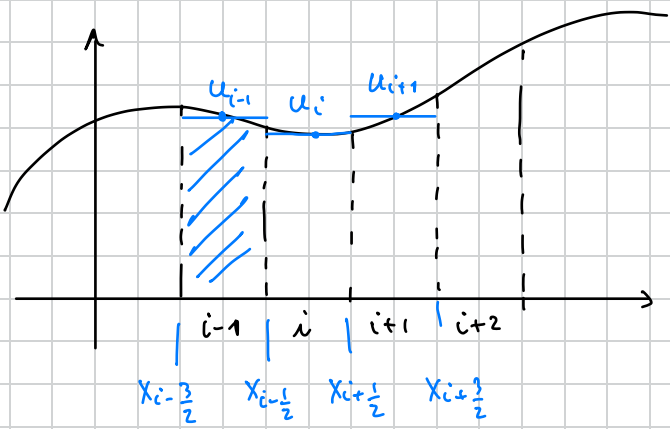
FINITE VOLUMES SCHEMES

- FE/FD ARE "NOT GOOD" CLOSE TO THE DISCONTINUITY
- CONSERVATIVE SCHEME AT THE DISCRETE LEVEL
- RELY ON THE INTEGRAL FORM

$$u_t + F(u)_x = 0 \qquad F \text{ non linear function of } u$$

$$\frac{d}{dt}\underbrace{\int_a^b u\,dx}_{\simeq \text{ AVERAGE OF } u \text{ OVER } (a,b)} = \underbrace{-F(u(b)) + F(u(a))}_{\text{CONSTANT W.R.T TIME}}$$



$$\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} u(x)\,dx = u_i \Delta x_i$$

- CELL
- GRID POINT

$$\Delta x_i = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}$$

$$\frac{d}{dt}\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} u\,dx = \underbrace{\frac{d}{dt}\left(u_i \Delta x_i\right) = F\left(u\left(x_{i-\frac{1}{2}},t\right)\right) - F\left(u\left(x_{i+\frac{1}{2}},t\right)\right)}$$

$$\int_{t_n}^{t_{n+1}} \frac{d}{dt}\left(u_i \Delta x_i\right) = \int_{t_n}^{t_{n+1}} \underbrace{F\left(u\left(x_{i-\frac{1}{2}},t\right)\right) - F\left(u\left(x_{i+\frac{1}{2}},t\right)\right)}\,dt$$

$$u_i(t_{n+1}) - u_i(t_n) = \frac{1}{\Delta x_i}\int_{t_n}^{t_{n+1}} \left(\quad\right)\,dt$$

$$F\left(u\left(x_{i-\frac{1}{2}},t\right)\right) \quad ?$$

FLUX RECONSTRUCTION

$$F\left(u\left(x_{i-\frac{1}{2}},t\right)\right) \simeq F_{i-\frac{1}{2}} \qquad \left(F(U_{j+1},U_j) \text{ FD}\right)$$

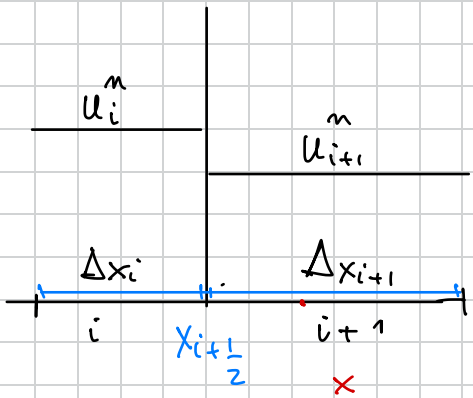
$$\frac{d}{dt}\underbrace{\sum_i u_i \Delta x_i}_{\Sigma \text{ CELL AVERAGES}} = \sum_i F_{i-\frac{1}{2}} - F_{i+\frac{1}{2}}$$

$$= F_{\frac{1}{2}} - \cancel{F_{3/2}} + \cancel{F_{3/2}} - \cancel{F_{5/2}} + \cancel{F_{5/2}} + \dots - F_{N+\frac{1}{2}}$$

$$= F_{1/2} - F_{N+\frac{1}{2}}$$

$\Rightarrow$ FV ARE CONSERVATIVE

- CENTRAL FLUX SCHEME



HOW TO COMPUTE

$u(x_{i+\frac{1}{2}}, t^n)$?

- USE A TAYLOR EXPANSION
- USE ONLY AVERAGE VALUES

$$x \in \left(x_{i+\frac{1}{2}},\ x_{i+\frac{1}{2}} + \Delta x_{i+1}\right)$$

$$u(x) = u\left(x_{i+\frac{1}{2}}\right) + \left(x - x_{i+\frac{1}{2}}\right)\frac{\partial u}{\partial x}\left(x_{i+\frac{1}{2}}\right) + \frac{\left(x - x_{i+\frac{1}{2}}\right)^2}{2}\frac{\partial^2 u}{\partial x^2}\left(x_{i+\frac{1}{2}}\right)$$

$$\underbrace{\frac{1}{\Delta x_{i+1}}\int_{x_{i+\frac{1}{2}}}^{x_{i+\frac{1}{2}}+\Delta x_{i+1}} u(x)\,dx}_{u_{i+1}} = u\left(x_{i+\frac{1}{2}}\right) + \frac{\Delta x_{i+1}}{2}\frac{\partial u}{\partial x}\left(x_{i+\frac{1}{2}}\right) + \frac{\Delta x_{i+1}^2}{6}\frac{\partial^2 u}{\partial x^2}\left(x_{i+\frac{1}{2}}\right)$$

$$u_{i+1} = u_{i+\frac{1}{2}} + \frac{\Delta x_{i+1}}{2}\frac{\partial u}{\partial x}\left(x_{i+\frac{1}{2}}\right) + \frac{\Delta x_{i+1}^2}{6}\frac{\partial^2 u}{\partial x^2}\left(x_{i+\frac{1}{2}}\right)$$

$$u_i = u_{i+\frac{1}{2}} - \frac{\Delta x_i}{2}\frac{\partial u}{\partial x}\left(x_{i+\frac{1}{2}}\right) + \frac{\Delta x_i^2}{6}\frac{\partial^2 u}{\partial x^2}\left(x_{i+\frac{1}{2}}\right)$$

$$\begin{cases} u_{i+\frac{1}{2}} = a\left(u_{i+1} - \frac{\Delta x_{i+1}}{2}\frac{\partial u}{\partial x} + \dots\right) \\ u_{i+\frac{1}{2}} = b\left(u_i + \frac{\Delta x_i}{2}\frac{\partial u}{\partial x} + \dots\right) \end{cases}$$

FIND a & b s.t. $u^+_{i+\frac{1}{2}} = u^-_{i+\frac{1}{2}}$

$$\begin{cases} a + b = 1 \\ b\frac{\Delta x_i}{2} - a\frac{\Delta x_{i+1}}{2} = 0 \end{cases}$$

$$a = 1 - b$$

$$b\frac{\Delta x_i}{2} - (1-b)\frac{\Delta x_{i+1}}{2} = 0$$

$$b\left(\frac{\Delta x_i + \Delta x_{i+1}}{2}\right) = \frac{\Delta x_{i+1}}{2}$$

$$b = \frac{\Delta x_{i+1}}{\Delta x_i + \Delta x_{i+1}} \qquad a = \frac{\Delta x_i}{\Delta x_i + \Delta x_{i+1}}$$

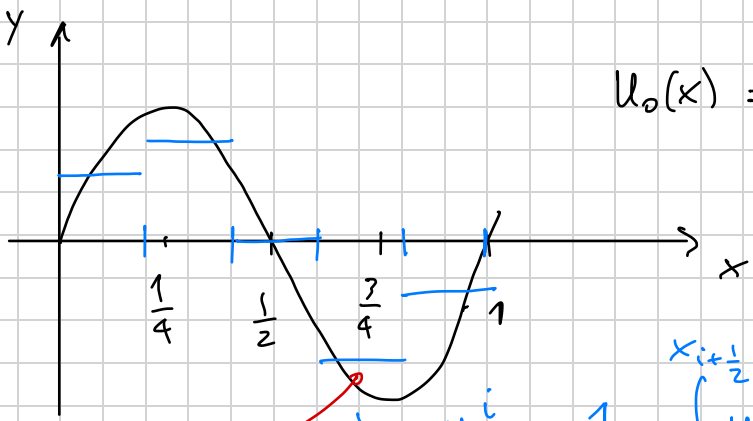
- $u_{i+\frac{1}{2}} = a\, u_{i+1} + b\, u_i$ WEIGHTED AVERAGE

if $\Delta x_i = \Delta x_{i+1}$ $\quad a = \frac{1}{2} = b$

$$u_{i+\frac{1}{2}} = \frac{1}{2} u_i + \frac{1}{2} u_{i+1}$$

- $F_{i+\frac{1}{2}} = F\left(\frac{1}{2} u_i + \frac{1}{2} u_{i+1}\right)$

$$\frac{d}{dt} u_i = \left(F_{i-\frac{1}{2}} + F_{i+\frac{1}{2}}\right) \frac{1}{\Delta x_i}$$



$u_0(x) = \sin(2\pi x)$

$$u_0^i = \frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} u_0(x)\, dx$$

$$\rightarrow u_0^i = \frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \sin(2\pi x)\,dx$$

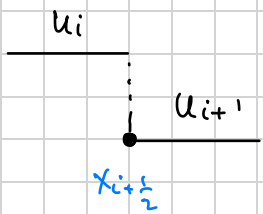
$$= -\frac{1}{2\pi} \frac{1}{\Delta x_i} \left(\cos\left(2\pi x_{i+\frac{1}{2}}\right) - \cos\left(2\pi x_{i-\frac{1}{2}}\right) \right)$$

⊕ 2ND ORDER ACCURATE (BY CONSTRUCTION)

⊖ SPURIOUS OSCILLATIONS CLOSE TO THE DISCONTINUITY

⊖ DOESN'T SATISFY PHYSICS

IDEA UPWIND



$$\sigma = \frac{[f]}{[u]}$$

$$\sigma > 0 \qquad f_{i+\frac{1}{2}} = f_i$$

$$\sigma < 0 \qquad f_{i+\frac{1}{2}} = f_{i+1}$$

$$u_{i+\frac{1}{2}} = \begin{cases} u_i & \text{if } \sigma > 0 \\ u_{i+1} & \text{if } \sigma < 0 \end{cases}$$

$$u_{i+\frac{1}{2}} = a\, u_i + b\, u_{i+1}$$

$\left(a = b = \frac{1}{2} \text{ for constant mesh siz} \right)$

⊖ 1ST ORDER SCHEME

* $a\,\Delta x_{i+1} - b\,\Delta x_i \neq 0$
- $a + b = 1$

⊕ CONSERVATIVE

⊖ $u_i \simeq u_{i+1}$ $\quad \sigma \simeq \infty$

$$\sigma = \frac{\partial f}{\partial u} \simeq u \qquad f(u) = \frac{u^2}{2}$$

GUDUNOV SCHEME (1959)

IDEA: U_j^n solution at time $t^n = n\Delta t$

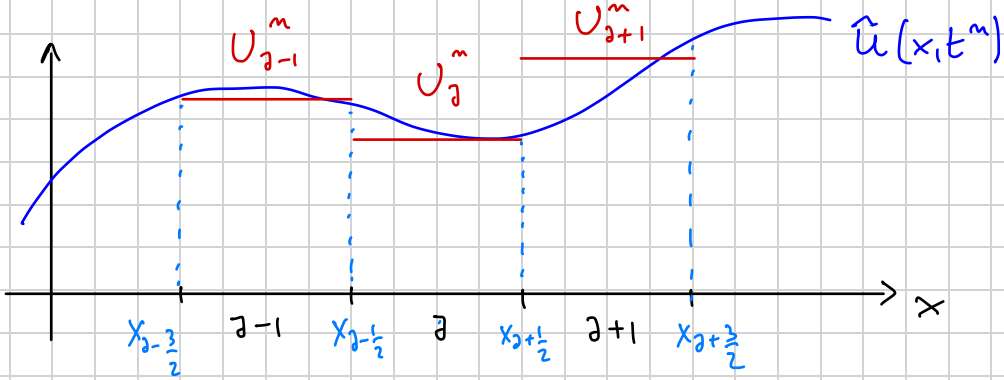
(piece-wise reconstruction)

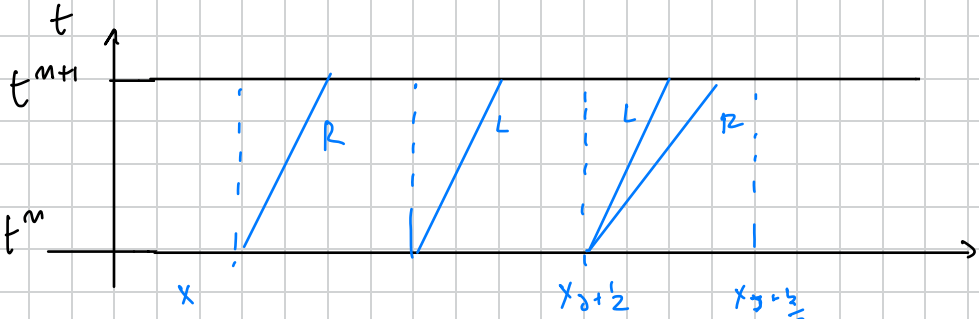
$$\tilde{u}(x, t^n) \simeq U_j^n \qquad x_{j-\frac{1}{2}} \leq x \leq x_{j+\frac{1}{2}}$$

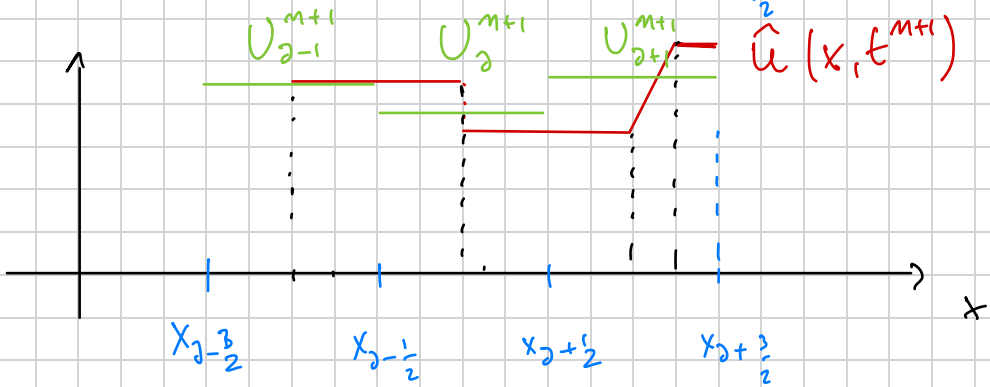
COMPUTE $\Rightarrow$ $\tilde{u}(x, t^{n+1})$ as the solution of the Riemann problem in $[t^n, t^{n+1}]$

PROJECT $\Rightarrow$

$$U_j^{n+1} = \frac{1}{\Delta x_j} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} \tilde{u}(x, t^{n+1})\, dx$$



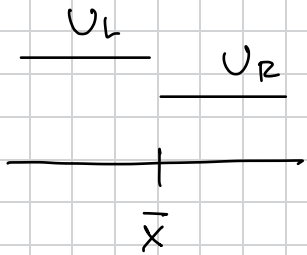




$$U_j^{n+1} = \frac{1}{\Delta x_j} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} \hat{u}(x,t^{n+1})\, dx$$

$$U_j^{n+1} = U_j^n + \frac{\Delta t}{\Delta x_j}\left[F(U_j^n, U_{j-1}^n) - F(U_{j+1}^n, U_j^n)\right]$$

$$F(U_L, U_R) = F(u^*(u_L, u_R))$$



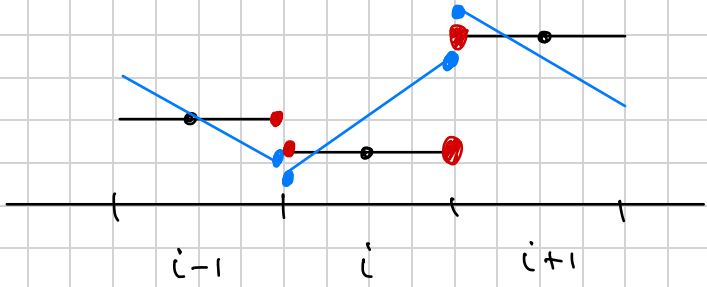
- U^* is defined in order to satisfy the entropy condition
- U^* solves the Riemann problem.

$$F(U_L, U_R) = \begin{cases} \min\limits_{u_L \le u \le u_R} F(u) & u_L < U_R \\ \max\limits_{u_R \le u \le u_L} F(u) & u_L > U_R \end{cases}$$

GODUNOV FLUX

⊖ 1ST ORDER ACCURATE (PIECE-WISE CONSTANT REC.)

⊕ HIGHER ACCURACY (by using high order reconstruction)



$$S_i = \frac{u_{i+1} - u_{i-1}}{2\Delta x}$$

S_i slope of the blue line

$$u^-_{i+\frac{1}{2}} = u_i + \frac{\Delta x}{2} S_{i-1}$$

$$u^+_{i+\frac{1}{2}} = u_i - \frac{\Delta x}{2} S_{i+1}$$

$$F_{i+\frac{1}{2}}\left(U^-_{i+\frac{1}{2}}, U^+_{i+\frac{1}{2}} \right.$$